Final Project - Lab: Working with the World Happiness Report in Tableau Public

Estimated time needed: 60 minutes

Required Software

You can complete this lab using the free Tableau Public online platform or the Tableau Desktop (Public Edition) app.

Prerequisites

You need a free Tableau Public account for this lab. If you haven't already done so, you can sign up for an account using the steps here

Data Set

The World Happiness Report is a landmark survey of the state of global happiness. The reports review the state of happiness in the world today and show how the new science of happiness explains personal and national variations in happiness.

This is a public data set available on the Kaggle website or World Happiness Report under the CC0: Public Domain license. The primary data set was retrieved from the World Happiness Report.

For this final project, you will work on the 2015-2023 data, which has been slightly modified for the purposes of this guided practice project.

Hold down the Ctrl key while you click to download it from here.

Attributes of this data set have been explained below:

Column	Description
country	The name of the country.
region	The geographic region or continent
happiness_score	A measure reflecting overall happiness.
gdp_per_capita	The extent to which GDP contributes to the calculation of the Happiness Score
social_support	A metric measuring social support.
healthy_life_expectancy	The extent to which life expectancy contributes to the calculation of the Happiness Score
freedom_to_make_life_choices	The extent to which freedom contributes to the calculation of the Happiness Score.
generosity	The extent to which generosity contributes to the calculation of the Happiness Score
year	The year of data collection, ranging from 2015 - 2023.

Task 1 - Happiness status overview

1.1 Create a highlighted table representing the average happiness score for each region with a relevant title and that exists as a filter for the dashboard:

- Title it "Average Happiness Score (Region)"
- Make it so the highlighted table highlights the important average happiness scores:

1.2 Create a map to display the 10 most happy countries.

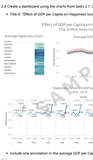
- Title it "Most Happy Countries"
- Make it so a parameter and use it as the filter for the *Country* data field
- Tooltips should include the name of the country, region, and average happiness score of the country
- Keep the map background as *Normal*
- Selected countries may be displayed with different colors on the map

1.3 Create a map to display the 10 least happy countries.

- Title it "Least Happy Countries"
- Make it so a parameter and use it as the filter for the *Country* data field
- Tooltips should include the name of the country, region, and average happiness score of the country
- Keep the map background as *Normal*
- Selected countries may be displayed with different colors on the map

1.4 Create a dashboard using the charts from tasks 1.1, 1.2, and 1.3.

- Title it "Happiness Data Overview (2015-2023)"
- Follow best practices on the dashboard. Use the same color, font, and font style for visualization titles. The white space between the visualizations should be approximately equal



Task 2 - Effect of GDP per capita on happiness score

2.1 Create a highlighted table to represent the average happiness score for the 10 happiest countries.

- Make use of a parameter as a filter to select the Top 10 Most Happy Countries
- Title it "Average Happiness Score"

2.2 Create a multiline chart to display the Average GDP per capita for the 10 happiest countries over the years.

- Make use of color to differentiate each country in the line graph
- Title it "Average GDP per Capita"

2.3 Create a line chart to display the happiness score for GDP per capita every year for the 10 happiest countries.

- Represent the happiness score in shades of blue by using the *Color* section of the Marks card
- Represent the 10 happiest countries by using a *Sub-Item Filter* for the *Happiness Score* field on the *Country* dimension
- Make use of a parameter as a filter to select the Top 10 Most Happy Countries
- Title it "Happiness Score Vs GDP per Capita (2015-2023)"

2.4 Create a dashboard using the charts from tasks 2.1, 2.2, and 2.3.

- Title it "Effect of GDP per Capita on Happiness Score (Top 10 Most Happy Countries)"

- Include one annotation in the average GDP per Capita chart as per your reference

reference annotation

- All the visualizations should represent the country selected by the user when clicked on in the "Average Happiness Score" table

Reference selection

- Follow best practices when building your dashboard. Use the same color, font, and font style for all visualization titles. The white space between the visualizations should be approximately equal

Task 3 - Effect of other factors on happiness score for the 10 happiest countries

3.1 Create a text table to display the happiness score of the 10 happiest countries for the years 2015 to 2023.

- Highlight the score with color
- Title it "Happiness Score of Most Happy Countries"

3.2 Create a dual combination bar and line chart to display freedom and social support for every year for the 10 happiest countries.

- Title it "Freedom and Social Support (Most Happy Countries)"

3.3 Create a dual combination bar and line chart to display generosity and healthy life expectancy for the 10 happiest countries.

- Title it "Generosity and Healthy Life Expectancy (Most Happy Countries)"

3.4 Create a dashboard including all three visualizations created in tasks 3.1, 3.2, and 3.3.

- Title it "Effect of Freedom, Social Support, Generosity and Healthy Life Expectancy (Most Happy Countries)"

- All the visualizations on the dashboard should represent the country selected by the user when clicked on in the "Happiness Score of Most Happy Countries" table

Task 4 - Effect of other factors on happiness score for the 10 least happy countries

- Repeat each part of the subtasks in Task 3 but create three visualizations for the "Least Happy Countries"
- Exclude two of the columns that do not have data for recent years when you create the "Happiness Score of Least Happy Countries" text table

- Title the dashboard "Effect of Freedom, Social Support, Generosity and Healthy Life Expectancy (Least Happy Countries)"

- All the visualizations on the dashboard should represent the country selected by the user when clicked on in the "Happiness Score of Least Happy Countries" table

Task 5 - Create a story and share your final project workbook

5.1 Create story points for each of the four dashboards you created in earlier tasks.

5.2 Provide relevant captions for each story point.

5.3 Include another story point captioned *Interviews*. Write three inferences you have gathered from the tasks above, which you would like to narrate while presenting the story to the stakeholders.

5.4 Publish your final project workbook and copy the link to it for submission.

To do this:

1. In the top right corner of the page, click **Publish As**. As **Publish A** (or as published this workbook already), then click **Publish As**, name your workbook *MyFinalProjectjoynames*, ensure you check the box **Save sheets as tabs**, and then click **Publish**.

   It is very important that you select the *Show sheets as tabs* checkbox otherwise it may be difficult for your peers to review your submission.

2. Select Authoring, click **File**, then click **Clone**.
3. In the Tableau Public web page for your final project story, click the **Share** icon in the top right toolbar above the final project workbook, and in the Share dialog box, click **Copy Link**. This will copy the workbook link to the clipboard.

Congratulations! You have completed the lab for this final project.

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Other Contributor(s)

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